

Received December 30, 2019, accepted January 20, 2020, date of publication January 30, 2020, date of current version February 10, 2020.

Digital Object Identifier 10.1109/ACCESS.2020.2970548

A Data-Driven Combined Algorithm for Abnormal Power Loss Detection in the Distribution Network

HUAN LONG^{1,2}, (Member, IEEE), CHANG CHEN¹, (Student Member, IEEE),
WEI GU¹, (Senior Member, IEEE), JIHUA XIE¹, ZHENG WANG³, AND GUODONG LI³

¹School of Electrical Engineering, Southeast University, Nanjing 210096, China

²Jiangsu Key Laboratory of Smart Grid Technology and Equipment, Nanjing 210096, China

³Tianjin Electric Power Research Institute, Tianjin 300041, China

Corresponding author: Huan Long (hlong@seu.edu.cn)

This work was supported by the State Grid Corporation of China Research Program: Technology and Application of Distribution Network Lean Line Loss Management Based on Big Data under Grant SGTJDK00DWJS1800014.

ABSTRACT Power loss, consisting of technical loss (TL) and non-technical loss (NTL), reflects the effective utilization rate of energy and the management level of power grids. This paper proposes a data-driven combined algorithm to systematically identify anomalies of power loss in the distribution network, including the abnormal type, time, and position. The detection process contains three stages: abnormal feeder detection, abnormal time detection, and abnormal position detection. The suspected abnormal feeders are first detected from all feeders in the distribution network by the data-driven algorithm based on the daily power supply and electricity sales data. Then, the control chart is employed to further monitor the fluctuation of the power loss of each suspected abnormal feeder and discover its abnormal time. Based on the detected abnormal time, its abnormal position is finally located through the risk assessment technology. Numerous experiments based on the real data show that the proposed data-driven combined algorithm can effectively detect and analyze abnormal power loss in the distribution network.

INDEX TERMS Power loss, abnormal detection, data-driven algorithm, control chart, risk assessment.

NOMENCLATURE

LCL, CL, UCL	Values of the lower control limit, center line and upper control limit of the control chart.	N_{OR}^i	Number of opening record of the electricity ammeter of the i -th feeder.
r	Power loss rate.	μ_X, σ_X	Range and mean of the X-bar control chart.
P_{sup}, P_{sales}	Quantity of power supply and electricity sales.	N_T	Number of the transformer branches in one suspected feeder.
$r_d^{i,t}$	Daily power loss rate of the i -th feeder in the t -th day.	δ_w	Risk index of TL, voltage and load, $w = \{TL, V, L\}$.
$\bar{r}_d^i$	Average daily power loss rate of the i -th feeder.	$S(\delta_w), S(\delta_w)_n$	Severity values of the risk index δ_w before and after normalization.
i, t	Index of the detected feeders and days.	L_a^j, L_p^j	Actual and predicted value of load of j -th transformer branch, $j = 1, \dots, N_T$.
N_f, N_d	Number of the detected feeders and days.	$r_{TL,a}^j, r_{TL,p}^j$	Actual and predicted value of TL of j -th transformer branch, $j = 1, \dots, N_T$.
CV_i	Coefficient of variation of daily power loss rate of the i -th feeder.	V_a^j, V_p^j	Actual and predicted value of voltage drop of j -th transformer branch, $j = 1, \dots, N_T$.
σ_i	Standard deviation of daily power loss rate of the i -th feeder.	S	The set of the related configuration parameters of the suspected feeder.
The associate editor coordinating the review of this manuscript and approving it for publication was Tariq Masood ¹ .		a, b, c, d	Parameters of the severity utility function of δ_w .

$T_{regular,w}, T_{caution,w}$	Threshold of each risk index for risk levels, Regular and Caution.
$N_{regular}, N_{caution}, N_{serious}$	Occurrence Number of each risk state based on all risk indexes.
$\lambda_{L1}, \lambda_{L2}$	The absolute value of the difference between the L_p and L_a .

I. INTRODUCTION

Power loss is an essential comprehensive technical and economic indicator for electric power companies. It consists of TL and NTL, reflecting the level of planning, design, production technology, and operation management of the power system. The TL is closely related to the grid equipment, topology, and operation mode. The NTL is caused by errors in metering equipment and poor management, such as electricity stealing and malfunctioning infrastructure [1], [2]. Thus, abnormal power loss detection in the distribution network is the necessity to improve the management level and ensure the economic benefit of electric power companies.

The current research works [1], [3]–[11], [13]–[17] on abnormal power loss detection in the distribution network are mainly focused on the NTL caused by electricity stealing behavior of low-voltage users. Relevant models have been established from three aspects: state-based, game theory-based, and data-driven-based [3]–[5]. The state-based methods [6], [7] identify the NTL according to the contradiction between the state estimation of the distribution network and the measurement data of power consumers, which are usually reliant on the detailed grid topology and measurement data of the distribution network. The game theory-based methods [8], [9] analyze the corresponding game equilibrium based on the behavioral interaction between abnormal power customers and electric power companies, which has only undergone theoretical derivation and simulation and has not yet been empirically tested.

With the development of smart grids, massive daily operation data are continuously generated, which provides a basis for data-driven detection on abnormal power loss. The data-driven-based methods achieve anomaly detection for power loss by analyzing the characteristics of the power consumers' load curves and the energy consumption distribution, which can be divided into three categories: regression-based, classification-based, and clustering-based [10]. These data-driven methods assume that the behavior patterns of abnormal consumption consumers will deviate from normal patterns, although the principles of these methods are different.

In [11], based on the electricity usage, and voltage data from smart meters, a physically inspired data-driven model was developed using a modified linear regression model to detect electricity theft by capturing the linear relationship between the power consumption and voltage magnitude of

customers on the same secondary. In [12], a method combined auto-regression and artificial neural networks, based on the data related to local weather, building usage and gas consumption, were presented for detecting wasting energy caused by anomalies in gas consumption. In [13], based on the historical customer consumption data, the support vector machine (SVM) extracted the load profile from the historical customer consumption data and detected the abnormal NTL according to the irregular consumption patterns of power customers. In [14], the optimum-path forest clustering algorithm was employed for unsupervised NTL recognition and anomaly detection problems. In [15], a multi-layer perceptron neural network was utilized as a classifier, which improved the performance on the NTL detection compared with the original error back-propagation algorithm. In [16], based on advanced metering infrastructure, a fraud detection system applying clustering algorithms, such as fuzzy clustering and K-means, was proposed to conduct abnormal power consumption detection by comparing the registered consumption reports before and after the discrepancy detection.

Three categories of the data-driven methods have their own characters for abnormal power loss detection. The regression-based methods need to establish the regression model for each consumer, which requires lots of data and computational resources. Although the classification-based methods have a high accuracy of anomaly detection, the classification-based methods require labeled abnormal power loss dataset, which is difficult to obtain [5]. On the contrary, the clustering-based methods dispense with the labeled dataset, but they have a strong dependence on parameter settings [10].

Most of the existing researches [4], [12], [13], [15]–[17] pay close attention to anomaly detection of low-voltage power consumption. Although the detection accuracy of these above methods, such as state-based and classification-based methods, is high, the high requirements for data are necessary. Besides, the abnormal TL and NTL are mixed in the real data, which cannot be directly distinguished, and the abnormal TL detection is ignored in most current research work. Furthermore, the anomaly detection for time and position of the suspected feeders is in the lack, which is also important information for the maintenance staff. As one of the statistical analysis methods to detect the abnormality, the theory of the control chart is mature, which can provide a statistically valid answer for anomaly detection [18]. The control chart generated from the historical data can analyze and judge the steady state of the operation process, which is suitable to identify the abnormal time of the suspected feeder. At present, the control chart is widely applied in production process control and product quality testing, while research in the abnormal power loss is relatively rare. In [1], a multivariate control chart was employed to detect abnormal NTL, establishing a reliable region for monitoring the measured variance. In this paper, abnormal position detection is based on the risk assessment technology of the distribution network. Risk assessment is an economic concept at the earliest. With the development of risk theory in power

systems and the practice in engineering applications, risk assessment has been gradually applied to various areas of the power industry, mainly including operational risk monitoring, equipment management, etc. [19]. In recent years, risk assessment technology has been employed in distributed power systems and active power distribution networks. In [20], based on the uncertainty theory, the risk assessment technology was utilized to sort and screen the fault risks in the distribution network to determine the risk with the highest level. In [21], a hierarchical risk assessment method was introduced to properly consider the impacts of active distribution network on risk analysis of the transmission system. However, at present, the idea of risk assessment is rarely used to locate the abnormal position of the feeders with abnormal power loss.

In order to solve the above problems, a novel data-driven combined algorithm for abnormal power loss detection is proposed in this paper, which is attributed to three stages: abnormal feeder detection, abnormal time detection, and abnormal position detection. Based on the daily power supply and electricity sales data, the suspected abnormal feeders existing abnormal TL and NTL are first detected by data-driven algorithms. To further identify the abnormal time, the control chart is utilized to analyze the daily power loss rate curve of each suspected abnormal feeder. On the basis of the feeder topology and load data of these suspected abnormal feeders, the abnormal position is detected by risk assessment technology during the detected abnormal time.

The remaining of this paper is as follows: Section II discusses the definition and types of abnormal TL and NTL, Section III describe the proposed data-driven combined algorithm, Section IV presents the performance of the proposed algorithm based on the real data from the Tianjin Electric Power Company, and the conclusion and future work are given in Section V.

II. DEFINITION AND TYPES OF POWER LOSS

The distribution network transmits the electricity to the end-users. In the transmission progress, the power loss is generated due to several reasons, such as impedance existed in the power equipment and management factor. The power loss rate, r , is determined by the power supply and electricity sales, and formulated as (1):

$$r = \frac{P_{sup} - P_{sales}}{P_{sup}} \times 100\% \quad (1)$$

where P_{sup} denotes the quantity of power supply, and P_{sales} represents the quantity of electricity sales [22]. Both too high and too low power loss rate are considered as abnormal power loss. The abnormal power loss can be attributed to abnormal TL and abnormal NTL. In the actual distribution network, some feeders both have abnormal TL and NTL. To cut down the power loss and improve the power transmission efficiency, it is necessary and important to identify the type of the abnormal power loss.

A. TL

The TL is the power loss caused by the normal power transmission and conversion, which can be calculated according to the parameters of power equipment, the current operation mode of the power system, the distribution of the power flow, and the load data. In the actual operation of the power grid, the too-high TL of the feeder is considered abnormal, which may be caused by the equipment with high power loss rate, small power supply wire radius, insufficient reactive power compensation, and so on.

B. NTL

The rest part of the total power loss, except TL, is called NTL, which is usually caused by the low management level. The electricity theft, defective meters, and a malfunctioning infrastructure are the major causes of abnormal NTL in the distribution network, especially electricity theft. The patterns of electricity theft behavior are elusive, such as modifying the electricity ammeters by tempering the internal data, connecting the public wire bypassing the electricity ammeters.

III. THE PROPOSED DATA-DRIVEN COMBINED ALGORITHM

In this section, the structure and process of the proposed data-driven combined algorithm are presented, and the details of each part are described later.

A. THE STRUCTURE OF THE PROPOSED ALGORITHM

This paper proposes a data-driven combined algorithm to systematically detect abnormal power loss in the distribution network, composed of abnormal feeder detection, abnormal time detection, and abnormal position detection. The structure of the proposed algorithm is presented in Fig. 1. In the stage of abnormal feeder detection, the abnormal feeder and corresponding abnormal type are identified based on the daily data of each feeder, P_{sup} , and P_{sales} . Considering that there are no labeled samples at the initial power loss detection, the clustering-based detection model is employed. With the continuous accumulation of the labeled samples, the classification-based detection model is utilized to improve the stability of abnormal feeder detection. The abnormal time detection is further performed for the suspected abnormal feeder. The upper control limit (UCL), center line (CL), and lower control limit (LCL) of the control chart is determined by the historical normal data. If the point of daily power loss rate, r_d , falls outside the control limits, the corresponding time is regarded as abnormal. Based on the detected abnormal time, the risk assessment technology is employed to pinpoint the specific abnormal position by calculating the risk indexes.

B. ABNORMAL FEEDER DETECTION

Generally speaking, the power loss features of abnormal feeders are different from those of normal ones. For feeders with

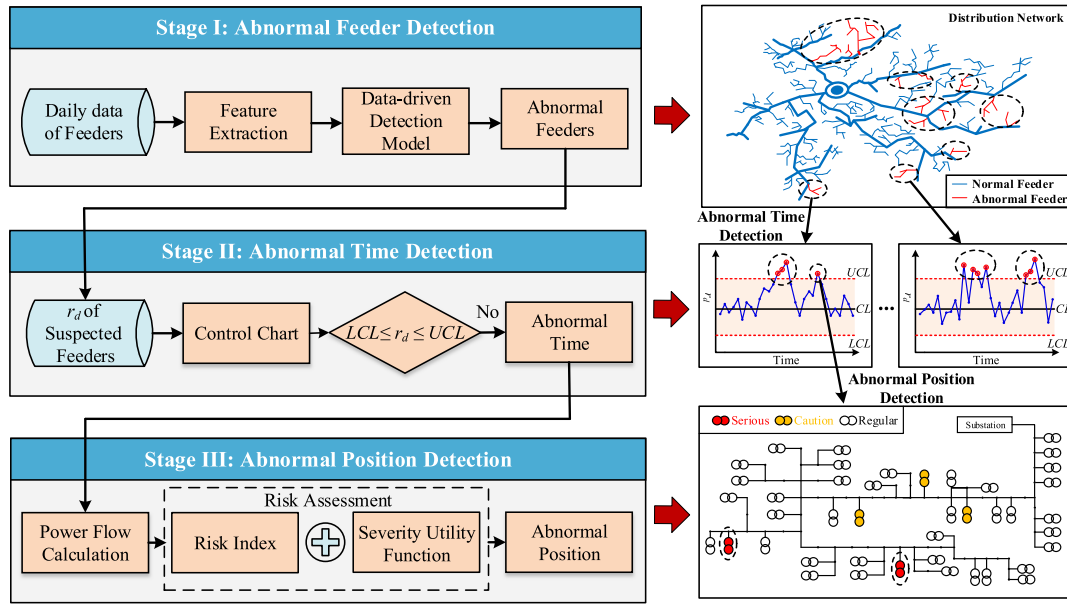


FIGURE 1. The flowchart and schematic diagram of the proposed algorithm.

abnormal TL, their daily power loss rate usually keeps a relatively high and stable value. For feeders with abnormal NTL, their fluctuation of daily power loss rate is more variable, and opening records of electricity ammeters may be larger due to the electricity theft by the ammeter.

The feature extraction directly impacts the performance of the anomaly detection for feeders. In this paper, three features related to the daily power loss rate and electricity ammeters are extracted as follows.

1) Average daily power loss rate: denote the daily power loss rate of the i -th feeder in the t -th day as $r_d^{i,t}$. N_f represents the number of the detected feeders, and N_d denotes the number of detected days. The average daily power loss rate of the i -th feeder is formulated as (2).

$$\bar{r}_d^i = \frac{1}{N_d} \sum_{t=1}^{N_d} r_d^{i,t}, \quad i = 1, 2, \dots, N_f \quad (2)$$

2) Coefficient of variation of daily power loss rate: it reflects the fluctuation of daily power loss rate. CV_i denotes the coefficient of variation of r_d^i in (3), and σ_i represents the standard deviation of the daily power loss rate of the i -th feeder in (4). The large the value of the coefficient of variation, the great fluctuation of the daily power loss rate.

$$CV_i = \frac{\sigma_i}{\bar{r}_d^i}, \quad i = 1, 2, \dots, N_f \quad (3)$$

$$\sigma_i = \sqrt{\frac{\sum_{t=1}^{N_d} (r_d^{i,t} - \bar{r}_d^i)^2}{N_d - 1}}, \quad i = 1, 2, \dots, N_f \quad (4)$$

3) Total opening records of electricity ammeters: generally speaking, the electricity ammeters are only opened in the

periodic maintenance. The electricity theft by the ammeters adds the number of opening the electricity ammeter. Thus, with the increase of opening records of electricity ammeters, the possibility of the illegal use of electricity rises. The total opening records of electricity ammeters of the i -th feeder, N_{OR}^i , is shown as (5). The $N_{OR}^{i,t}$ denotes the opening records of electricity ammeters of the i -th feeder on the t -th day.

$$N_{OR}^i = \sum_{t=1}^{N_d} N_{OR}^{i,t}, \quad i = 1, 2, \dots, N_f \quad (5)$$

The process of abnormal feeder detection based on the data-driven model is presented in Fig. 2. The data-driven model includes the clustering-based model and classification-based model. In the early detection stage, the normal and abnormal labels of the dataset are absence and unknown. In this situation, the clustering-based detection model is applied to conduct the abnormal feeder detection, and the detection process is marked by the black line in Fig. 2. After detecting the abnormal feeders, the grid staff is required to inspect and confirm on site. In this way, the unlabeled training set can be labeled as normal, abnormal TL, and abnormal NTL. When the labeled training set is large enough, the classification-based detection model is instead of the clustering-based one. The detection process of the classification based model is marked by the red line in Fig. 2. Compared with the clustering-based model, the classification-based model is more stable, and its detection result is less impacted by the training dataset. The more important point is that the classification-based model can detect the abnormal type, while the clustering-based model only detects the abnormal feeders.

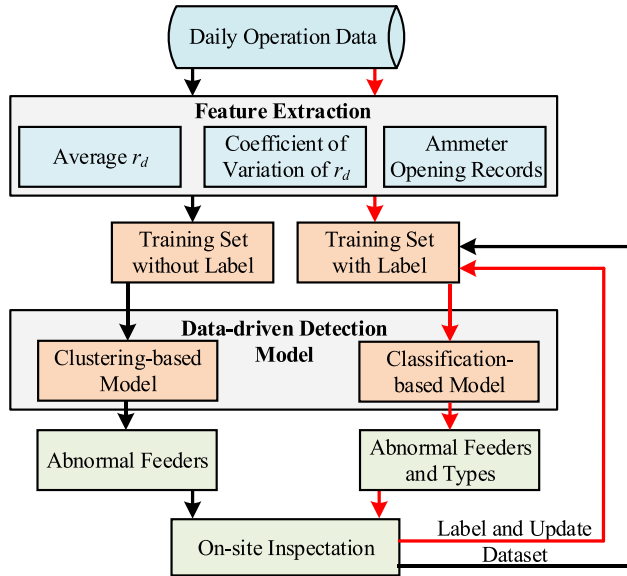


FIGURE 2. The process of the data-driven abnormal feeder detection model algorithm.

C. ABNORMAL TIME DETECTION

In this stage, the control chart is employed to monitor the fluctuation of r_d of the suspected abnormal feeders and discover their abnormal time. The control chart is a scientifically designed diagram with control limits to analyze and judge whether the process is in a stable state and to distinguish normal and abnormal fluctuations. In Fig. 1, the UCL and LCL are usually set at the position of ± 3 standard deviation, and CL is the average of the controlled statistics. If the point falls outside the UCL and LCL , its corresponding state is usually regarded as abnormal [23].

In this paper, the X bar control chart is utilized due to its high sensitivity. It detects the abnormal based on the average values of data. The accidental fluctuation of r_d is a slight deviation through the average effect of the mean, while the abnormal fluctuation of r_d usually deviates in the same direction to cause great deviation. Thus, the X bar control chart is suitable for abnormal time detection.

A pseudo-code of determining the control limits of the X bar control chart is provided in Algorithm 1. Since the load level different with the month greatly impacts the power loss rate, the whole data are divided into different N subgroups based on the date information which contains M days, denoted as $\mathbf{D}_{n \times m}$, $m = 1, 2, \dots, M$, and $n = 1, 2, \dots, N$.

Firstly, the range and average of each subgroup are calculated as $Sub_Range[n]$ and $Sub_Mean[n]$ (Line 1 to Line 4). The functions $\max()$, $\min()$, and $\text{mean}()$ are respectively to obtain the maximum value, minimum value and mean value of the dataset. Secondly, the mean range and average of all subgroups are computed as σ_X and μ_X (Line 5 to Line 6). Finally, the control limits of the X bar control chart, UCL , LCL , and CL are obtained (Line 7 to Line 10), where C_M is a constant value related to M and can be inquired in the control chart coefficient table [24].

Algorithm 1 X bar Control Chart

Input: Given Dataset $\mathbf{D} = \{\mathbf{D}_{n \times m}, m = 1, 2, \dots, M, n = 1, 2, \dots, N\}$

Output: UCL, CL, LCL

```

1:   for  $n \leftarrow 1$  to  $N$  do
2:        $Sub\_Range[n] \leftarrow \max(\mathbf{D}_{n \times m}) - \min(\mathbf{D}_{n \times m})$ ;
3:        $Sub\_Mean[n] \leftarrow \text{mean}(\mathbf{D}_{n \times m})$ ;
4:   end for
5:    $\sigma_X \leftarrow \frac{1}{N} \sum_{n=1}^N Sub\_Range[n]$ ;
6:    $\mu_X \leftarrow \frac{1}{N} \sum_{n=1}^N Sub\_Mean[n]$ ;
7:    $UCL \leftarrow \mu_X + C_M \cdot \sigma_X$ ;
8:    $CL \leftarrow \mu_X$ ;
9:    $LCL \leftarrow \mu_X - C_M \cdot \sigma_X$ ;
10:  return  $UCL, CL, LCL$ .
  
```

D. ABNORMAL POSITION DETECTION

In order to locate the specific abnormal position of the suspected feeders during the detected abnormal time, the abnormality severity risk level of each transformer of the suspected feeders is assessed.

In the current risk assessment of the distribution network, the preset accidents are regarded as the starting point, and the risk index is established to assess their occurrence probability and the consequential loss. To quantify the severity of the consequential loss caused by these accidents and embody them in risk index, a severity function is built based on the utility theory to reflect the satisfaction of terminal users or grid operators. The abnormal power loss in the distribution network is mainly caused by the abnormal power consumption behavior and instant faults of the power equipment and so on, which will be reflected in the voltage, TL, and electricity sales.

This paper compares the deviation between the actual value in the actual operation and the reference value of the load, voltage, and TL under the given normal situation. Thus, three risk indexes are constructed, the load offset, δ_L , the TL rate offset of each transformer branch in the feeder, δ_{TL} , and the voltage offset, δ_V . Since the load greatly impacts the value of the power loss, the accuracy of the risk assessment results cannot be guaranteed if the reference value is determined only by the experience of the grid staff. Thus, the future trend of power flow is considered as the reference value under normal circumstances and calculated based on the predicted load.

In one suspected feeder, suppose N_T transformers exist. Denote the actual and predicted value of the load of the j -th transformer branch as L_a^j and L_p^j , $j = 1, \dots, N_T$. Through the power flow calculation, the actual and predicted value of the TL rate and voltage drop of each transformer branch are obtained as $r_{TL,a}^j, r_{TL,p}^j, V_a^j$, and V_p^j , shown as (6). The function $f(\bullet)$ denotes power flow calculation, and $\mathbf{S}$ is the set of the related configuration parameters of the suspected feeder, including the feeder topology, power equipment parameters. The three risk indexes of the j -th transformer branch, δ_w^j ,

$w = \{TL, V, L\}, j = 1, \dots, N_T$, are expressed in (7).

$$[\mathbf{r}_{TL,a}, \mathbf{V}_a] = f(\mathbf{L}_a, \mathbf{S}), [\mathbf{r}_{TL,p}, \mathbf{V}_p] = f(\mathbf{L}_p, \mathbf{S}) \quad (6)$$

$$\delta_{TL}^j = \frac{|r_{TL,a}^j - r_{TL,p}^j|}{r_{TL,a}^j}, \quad \delta_V^j = \frac{|V_a^j - V_p^j|}{V_a^j}, \quad \delta_L^j = \frac{|L_a^j - L_p^j|}{L_a^j} \quad (7)$$

In order to measure the risk severity of the risk indexes δ_w , the severity utility function is utilized as $S(\delta_w)$. Both first derivative $S'(\delta_w)$ and second derivative $S''(\delta_w)$ exist and satisfy $S'(\delta_w) > 0$ and $S''(\delta_w) > 0$. $S'(\delta_w)$ stands for that the degree of abnormality severity goes up with the increase of δ_w . The term $S''(\delta_w) > 0$ indicates that the increasing degree of abnormality severity becomes more and more fast with the rise of the δ_w . The severity utility function of δ_w is shown in (8), where a, b, c, d are positive parameters [25]. In this paper, a, c , and d are set to 1, b equals 0. The simplified severity utility function is presented in (9). In order to eliminate the influence of different dimensions of the risk indexes, normalization of the severity utility function is required and denoted as $S(\delta_w)_n$. The value of $S(\delta_w)_n$ close to 1 means the high level of the risk severity.

$$S(\delta_w) = \frac{1}{c}(e^{a(\delta_w+b)} - d), \quad w = \{TL, V, L\} \quad (8)$$

$$S(\delta_w) = e^{\delta_w} - 1, \quad w = \{TL, V, L\} \quad (9)$$

To comprehensively analyze the three risk indexes and classify the risk severity level, the risk level is categorized into Serious, Caution, and Regular. For each risk index δ_w , the $S(\delta_w)_n$ is divided into three intervals, $[0, T_{regular,w})$, $[T_{regular,w}, T_{caution,w})$, $[T_{caution,w}, 1]$. The final risk level is determined by the number of $S(\delta_w)_n, w = \{TL, V, L\}$, located in the different interval, $N_{regular}, N_{caution}$, and $N_{serious}$, shown as (10)-(12).

$$N_{regular} = \sum_{w=\{TL, V, L\}} 1_{[0, T_{regular,w})}(S(\delta_w)_n) \quad (10)$$

$$N_{caution} = \sum_{w=\{TL, V, L\}} 1_{[T_{regular,w}, T_{caution,w})}(S(\delta_w)_n) \quad (11)$$

$$N_{serious} = \sum_{w=\{TL, V, L\}} 1_{[T_{caution,w}, 1]}(S(\delta_w)_n) \quad (12)$$

IV. CASE STUDY

In the case study part, the real data used in the experiments is first to describe. The detection performance of abnormal feeder detection, abnormal time detection, and abnormal position detection are displayed in the rest parts. The simulations were conducted with MATLAB and Python in a PC laptop with a 2.50-GHz Intel Core i5 processor and 8 GB of memory.

A. DATA DESCRIPTION

The utilized dataset in this paper was collected from the Tianjin Electric Power Company in China. The 10kV

TABLE 1. Modification principle of six abnormal patterns.

Pattern	Modification ($t = 1, 2, \dots, 24$)	Probability
1	$\tilde{L}_t = \alpha L_t$	0.2
2	$\tilde{L}_t = L_t - 1_{(st, et)}(t) L_t$ $st = \text{random}(0, 23 - \min_t)$, $et = st + \text{random}(\min_t, 24)$	0.05
3	$\tilde{L}_t = \beta_t L_t$	0.5
4	$\tilde{L}_t = \beta_t \times \text{mean}(L)$	0.1
5	$\tilde{L}_t = \text{mean}(L)$	0.1
6	$\tilde{L}_t = L_{24-t}$	0.05

TABLE 2. The detailed number of each type.

Label	Training set	Test set
-1	31	8
0	277	69
1	60	15

TABLE 3. The result of feature extraction after normalization.

i	$\tilde{r}_d$	CV_i	N_{OR}^i	Label
1	0.119	0.198	0	0
2	0.069	0.129	0	0
3	0.726	0.077	0	-1
4	0.329	0.659	1	+1
...	...	...	...	...

distribution network is regarded as the detection objective. The 460 feeders are selected from the 10kV distribution network to be detected. Since the data privacy of the private transformers, their data fail to be collected and not be considered in this paper. The data, covering from February 1st, 2017 to April 30th, 2019, include the topology parameters, the parameters of power equipment, load, electricity sales, opening records of the electricity ammeter.

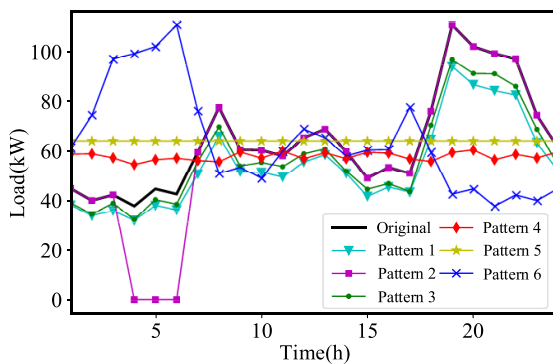
Given the records of the Tianjin Electric Power Company, the selected feeders operate stably and safely without abnormal NTL in this period. The 39 feeders out of the selected ones are marked as abnormal TL samples by the grid operation staff. To validate the performance of detecting the abnormal NTL, the 75 feeders are randomly chosen as abnormal NTL samples, and each feeder randomly modifies the daily load of 1~4 randomly selected transformers. To simulate the abnormal NTL, six abnormal patterns are chosen [4]. Table 1 presents the modification principle of each abnormal pattern. $\tilde{L}_t$ represents the modified daily load data. In pattern 2, pattern 3, and pattern 4, the coefficient parameters α and β_t , $t = 1, 2, \dots, 24$, are randomly generated in $[0.88, 0.98]$, and the $\min_t$ is set as 4 in this paper. An example of the abnormal patterns applied to a transform is shown in Fig. 3.

B. DETECTION RESULT OF ABNORMAL FEEDERS

To evaluate the data-driven detection model, the samples, including the normal and abnormal ones, are labeled (−1: abnormal TL, 0: normal, +1: abnormal NTL). All the samples

TABLE 4. Detection results of test set using the selected algorithms.

Algorithm	Cluster/Label	True value			Abnormal feeders		Abnormal TL		Abnormal NTL	
		0	-1	+1	DR(%)	FAR(%)	DR(%)	FAR(%)	DR(%)	FAR(%)
K-means	$C_1 : 0$	69	1	4	78.26	0	100	45.45	54.55	0
	$C_2 : -1$	0	7	5						
	$C_3 : +1$	0	0	6						
Spectral clustering	$C_1 : 0$	69	2	4	73.91	0	100	45.45	54.55	0
	$C_2 : -1$	0	6	5						
	$C_3 : +1$	0	0	6						
GMM	$C_1 : 0$	69	1	1	91.30	0	100	42.86	57.14	0
	$C_2 : -1$	0	7	6						
	$C_3 : +1$	0	0	8						
SVM	0	69	0	1	95.65	0	100	35.71	64.29	0
	-1	0	8	5						
	+1	0	0	9						

**FIGURE 3.** Abnormal load curves resulted from six types of abnormal patterns.

are randomly assigned to the training set and test set, shown in Table 2.

The evaluation criterion utilized in this paper is denoted the detection rate, DR , and the false alarm rate, FAR , as (13)-(14).

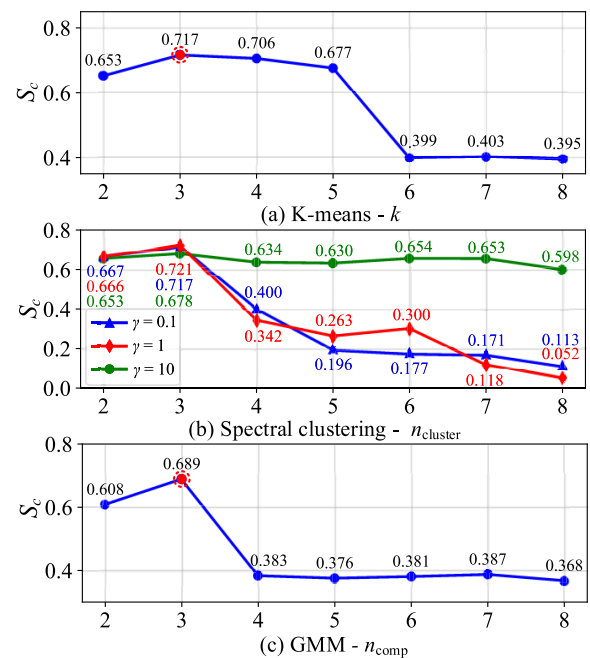
$$DR = \frac{N_{TA}}{N_{TA} + N_{FA}} \times 100\% \quad (13)$$

$$FAR = \frac{N_{FN}}{N_{TN} + N_{FN}} \times 100\% \quad (14)$$

where N_{TA} , N_{FA} , N_{TN} , and N_{FN} represent the number of abnormal samples detected as abnormal, abnormal samples detected as normal, normal samples detected as normal, and normal samples detected as abnormal, respectively.

Clustering algorithms are mainly divided into the centroid-based, spectral-based, density-based, and distribution-based. In this paper, the clustering algorithms, K-means [26], spectral clustering [27], and GMM [28], and classification algorithm, SVM [29], are utilized to detect abnormal feeders. The result of the extracted features after normalization is shown in Table 3.

The optimal parameter setting of the clustering algorithms is determined by the silhouette coefficient, S_c . The S_c is closer to 1, the better the clustering result is. The selection result of parameter settings for different algorithms is shown as Fig. 4. It is evident that K-means obtains the best clustering results when the number of clusters, k , equals 3. In spectral

**FIGURE 4.** The results of S_c under different parameter settings in the clustering algorithms.**TABLE 5.** The detailed anomalies of the feeder in fig. 5.

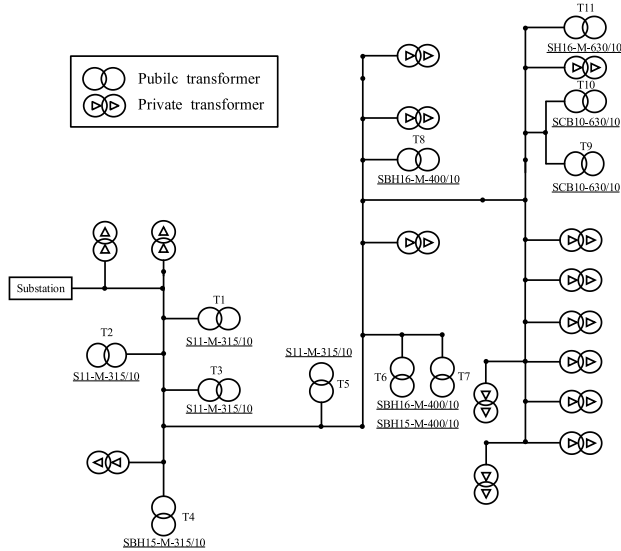
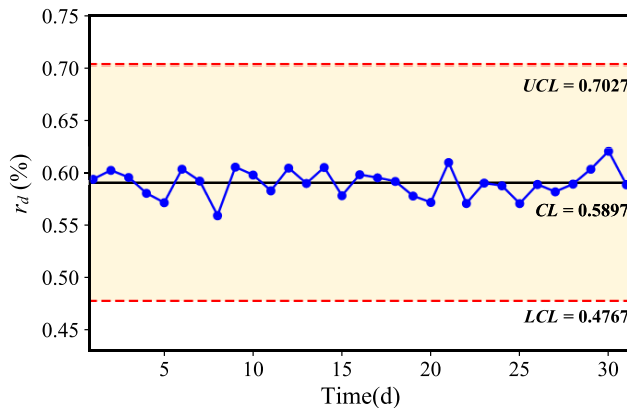
Anomaly	Abnormal Time	Abnormal Position	Abnormal Pattern
1	March 8th, 2019 ~ March 14th, 2019	T2	3
2	March 22th, 2019 ~ March 28th, 2019	T4 T6 T9	3 1 1

clustering, the number of clusters, $n_{cluster}$, and kernel coefficient, γ are set to 1 and 3, respectively, to achieve the max value of the S_c . In GMM, the optimal number of components, n_{comp} , equals 3. In SVM, radial basis function (RBF) is selected as the kernel. The grid search [29] is applied to search the optimal kernel parameter, γ , and the penalty for the error term, C_p , which are set to 1 and 1, respectively.

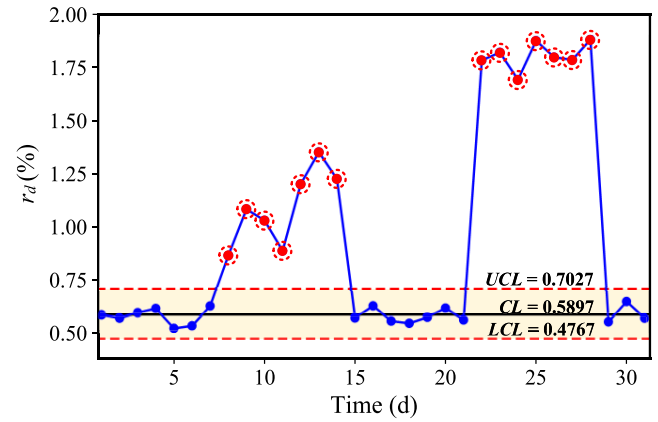
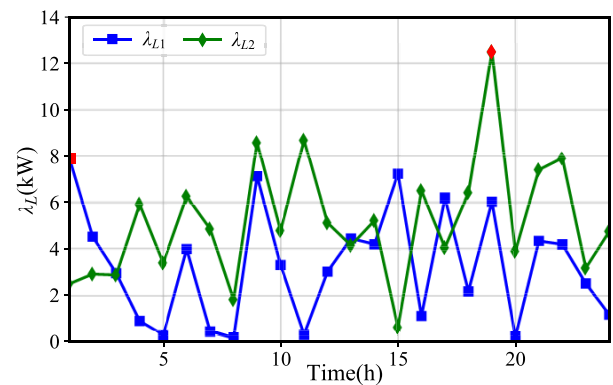
Based on the above parameter settings, the detection results of abnormal feeders using different data-driven models are

TABLE 6. Prediction result of 11 public transformers.

Public Transformer	MAPE(%)	Public Transformer	MAPE(%)
T1	2.5123	T7	2.4949
T2	2.5410	T8	2.5411
T3	2.4818	T9	2.4771
T4	2.6326	T10	2.4387
T5	2.5093	T11	2.4824
T6	2.4851		

**FIGURE 5.** The selected feeder topology.**FIGURE 6.** The X bar control chart for analysis.

presented in Table 4. Since the distribution network operates stably and safely, it is reasonable to assume that most of the selected feeders are normal. Thus, in the detection result of the clustering-based model, the clusters with small and large samples are separately regarded as abnormal and normal. Taking the K-means algorithm as an example, Cluster C_1 is regarded as normal, and Cluster C_2 and Cluster C_3 are regarded as abnormal. Compared with the clustering models, the classification models can directly identify the normal,

**FIGURE 7.** The X bar control chart for abnormal time detection.**FIGURE 8.** λ_{L1} and λ_{L2} of Anomaly Case 1 and 2.

abnormal TL and abnormal NTL samples by taking advantage of the labeled samples. To compare algorithms' performances, the detected abnormal clusters of clustering-based algorithms is further marked. It is assumed that the label of each cluster is determined by the label with the maximum number in this cluster. The detailed result is shown as Table 4.

As observed in Table 4, GMM performs the best among all the selected clustering algorithms, and the performance of the classification model is better than the clustering ones. Besides, the classifier of the classification algorithm can be trained offline, which makes anomaly detection of feeders more efficient. However, it is clear that some feeders are still detected incorrectly. There are existed some reasons: a) the $\bar{r}_d^i$ of some actual abnormal TL feeders is at a medium level, which is close to the actual normal feeders; b) the abnormal NTL caused by minor abnormal electricity behavior of power consumers is not apparent, as a result of which extracted features are close to the normal samples; c) the characters of some actual abnormal NTL feeders are similar as the abnormal TL samples and detected as abnormal TL.

C. DETECTION RESULT OF ABNORMAL TIME

The X bar control chart of each suspected feeder is established to analyze the fluctuation of r_d and detect the specific abnormal time. In this paper, the data in 2017 and 2018 are

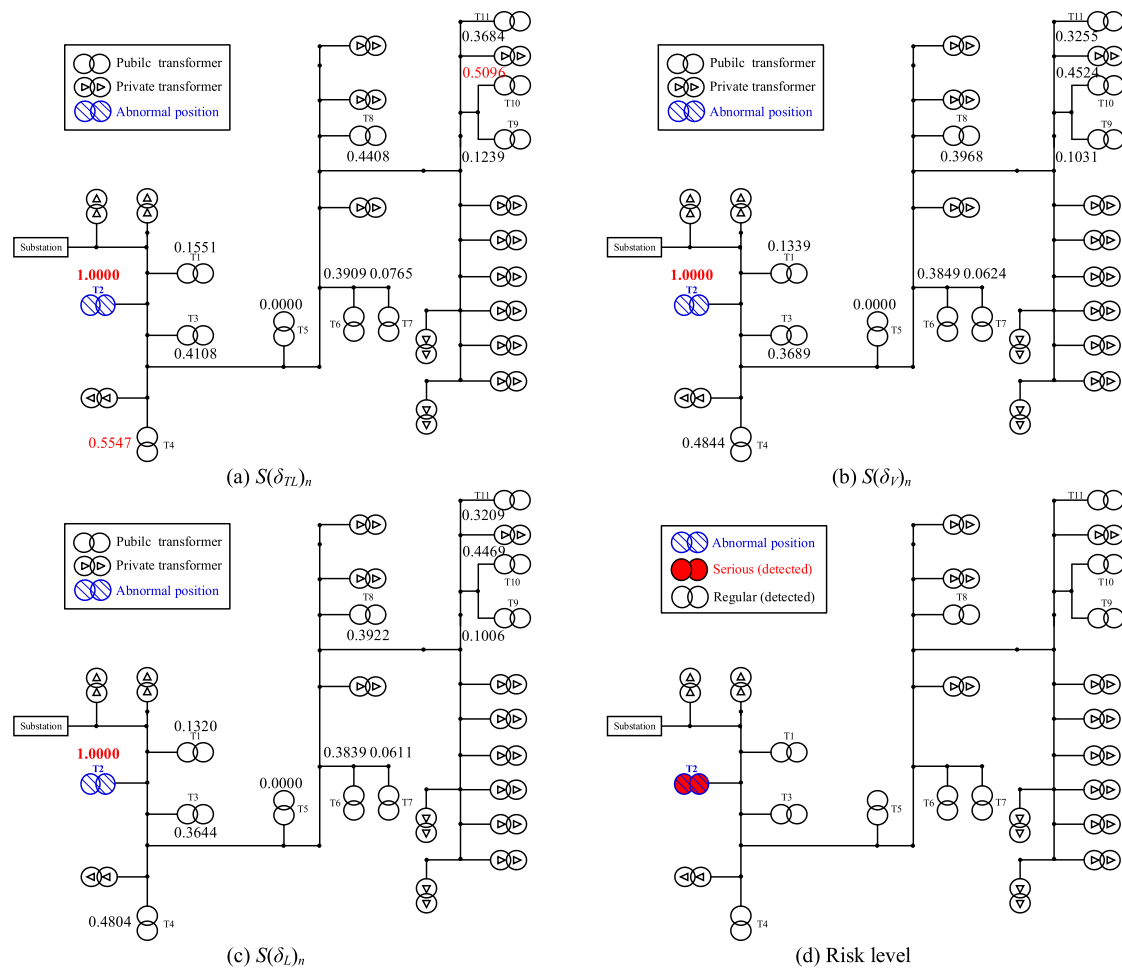
FIGURE 9. The $S(\delta_w)_n$ and risk level in Anomaly Case 1.

TABLE 7. Risk level in anomaly case 1.

	$S(\delta_{TL})_n$	$S(\delta_V)_n$	$S(\delta_L)_n$	$N_{serious}$	$N_{caution}$	$N_{regular}$	Risk Level
T1	0.1551	0.1339	0.1320	0	0	3	Regular
T2	1.0000	1.0000	1.0000	3	0	0	Serious
T3	0.4108	0.3689	0.3644	0	0	3	Regular
T4	0.5547	0.4844	0.4804	0	1	2	Regular
T5	0.0000	0.0000	0.0000	0	0	3	Regular
T6	0.3909	0.3849	0.3839	0	0	3	Regular
T7	0.0765	0.0624	0.0611	0	0	3	Regular
T8	0.4408	0.3968	0.3922	0	0	3	Regular
T9	0.1239	0.1031	0.1006	0	0	3	Regular
T10	0.5096	0.4524	0.4469	0	1	2	Regular
T11	0.3684	0.3255	0.3209	0	0	3	Regular

utilized to obtain the UCL , CL , and LCL of the control chart, and data from March 1st, 2019 to March 31st, 2019, are tested. To display the performance of abnormal time detection, one of the abnormal NTL samples is taken as an example. The power grid topology of this selected feeder is depicted in Fig. 5, containing 11 public transformers and 15 private transformers.

According to the generation records of abnormal NTL samples, the detailed anomalies of this feeder are listed in Table 5. There are two anomalies separately happened in two different times on different transformers with different abnormal patterns.

Fig. 6 shows the $\bar{X}$ control chart established based on the historical r_d in 2017 and 2018. All data fall within

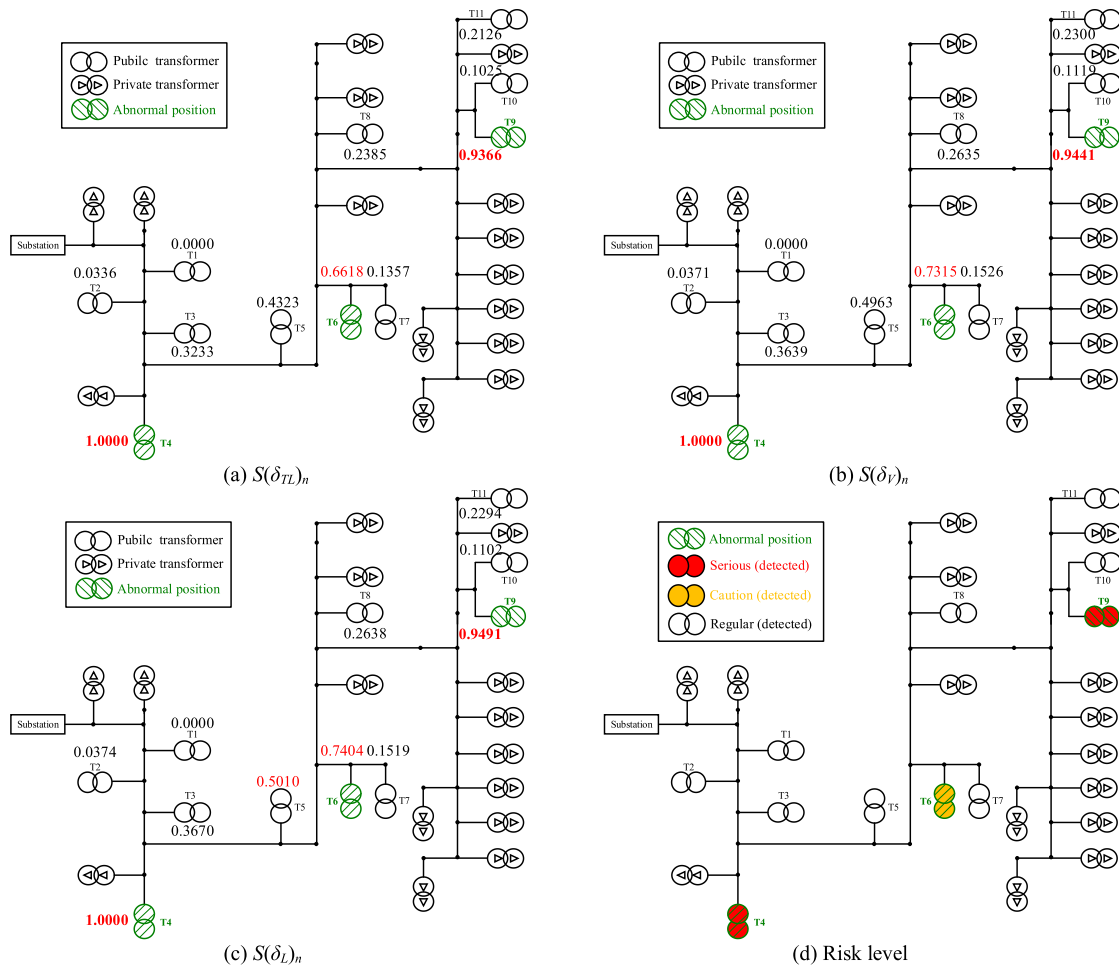


FIGURE 10. The $S(>w)_n$ and risk level in Anomaly Case 2.

TABLE 8. Risk level in anomaly case 2.

	$S(\delta_{TL})_n$	$S(\delta_V)_n$	$S(\delta_L)_n$	$N_{serious}$	$N_{caution}$	$N_{regular}$	Risk Level
T1	0.0000	0.0000	0.0000	0	0	3	Regular
T2	0.0336	0.0374	0.0371	0	0	3	Regular
T3	0.3233	0.3670	0.3639	0	0	3	Regular
T4	1.0000	1.0000	1.0000	3	0	0	Serious
T5	0.4323	0.5010	0.4963	0	1	2	Regular
T6	0.6618	0.7404	0.7315	0	3	0	Caution
T7	0.1357	0.1519	0.1526	0	0	3	Regular
T8	0.2385	0.2638	0.2635	0	0	3	Regular
T9	0.2385	0.2638	0.2635	3	0	0	Serious
T10	0.1025	0.1102	0.1119	0	0	3	Regular
T11	0.2126	0.2294	0.2300	0	0	3	Regular

the control limits, which suggests that the $\bar{X}$ control chart achieves a steady state. The UCL , CL , and LCL of the $\bar{X}$ control chart in this paper are calculated as 0.7207, 0.5897, and 0.4767, respectively. Based on the control limits of Fig. 6, anomaly detection for time can be conducted, as shown in Fig. 7. It is evident that there are 14 points marked as red locating outside the UCL , which is corresponding to

the detailed anomalies listed in Table 5. The DR and FAR of abnormal time detection is 100% and 0%, respectively.

D. DETECTION RESULT OF ABNORMAL POSITION

To detect the abnormal position in Anomaly 1 and Anomaly 2, respectively, the risk assessment is employed to analyze the risk state of each transformer branch during

TABLE 9. The L_a , γ_L and δ_L of T4, T6, and T9.

	$L_a(\text{kW})$	$\gamma_L(\text{kW})$	δ_L
T4	62.7540	5.9357	0.0946
T6	21.6932	1.5402	0.0710
T9	19.6073	1.7648	0.0900

the abnormal time. In the real application, the electric power company staff can conduct anomaly detection for the position by the hour, which enables them to further determine the specific abnormal time and location.

In this paper, $T_{\text{regular},w}$, $T_{\text{caution},w}$ are set as 0.5 and 0.8, respectively. If $N_{\text{serious}} > 0$, the risk level of the transformer branch is marked as Serious. If $N_{\text{serious}} = 0$ and $N_{\text{caution}} \geq 2$, the risk level of the transformer branch is marked as Caution. Otherwise, the risk level is marked as regular.

Due to the L_p directly impacting the risk assessment results of abnormal position detection, it is of great importance to ensure the load prediction accuracy, which is evaluated by mean absolute percentage error (MAPE): The difference between the original value of the actual load and L_p is divided by the original value of the actual load; The absolute value in this calculation is summed for every predicted point in time and divided by the number of fitted points. The one-step-ahead predicted load data are obtained every 15 minutes. The MAPE of the public transformers in the selected feeder is shown in Table 6.

To present the risk assessment results in this paper, the representative day and hour are selected according to Fig. 7 and Fig. 8, respectively. In Fig. 7, the two abnormal days, March 8th and March 24th, are chosen because their r_d are smallest during each abnormal time of two anomalies. Fig. 8 presents the absolute value of the difference between the L_p and L_a of each hour of March 8th and March 24th, expressed as λ_{L1} and λ_{L2} . It is obvious that the λ_{L1} achieves the highest value at the 1st hour on March 8th, and λ_{L2} achieves the highest value at the 19th hour on March 24th, which indicates that the abnormal NTL is considered as the severest at this time. The abnormal position is further detected of the 1st hour on March 8th and the 19th hour on March 24th, which is denoted as Anomaly Case 1 and 2, respectively.

1) ANOMALY CASE 1

In Anomaly Case 1, the abnormal position detection of the selected feeder in Fig. 5 is conducted at the 1st hour on March 8th. At this time, the $S(\delta_{TL})_n$, $S(\delta_V)_n$, and $S(\delta_L)_n$ of each transformer are displayed in Fig. 9(a)-(c) and their ranges are presented in Table 7. The risk level of each transformer in Anomaly Case 1 can be obtained according to the proposed rules, depicted in Fig. 9(d) and Table 7. As observed in Fig. 9(d), it is obvious that the risk level of T2 is detected as Serious, which is in accordance with the known abnormal position in Anomaly Case 1.

2) ANOMALY CASE 2

In Anomaly Case 2, the abnormal position detection is conducted at the 19th hour on March 24th. At this time,

the $S(\delta_{TL})_n$, $S(\delta_V)_n$, and $S(\delta_L)_n$ of each transformer are displayed in Fig. 10(a)-(c) and their ranges are shown in Table 8.

The risk level of each transformer in Anomaly Case 2 can be obtained according to the proposed rules shown in Table 2. As presented in Fig. 10(d), it is evident that the risk level of T4, T6 and T9 are separately detected as Serious, Caution, and Serious, respectively, which corresponds to the known abnormal position in Anomaly Case 2. From Table 9 we can see that the higher the δ_L of the transformer, the higher its risk level.

V. CONCLUSION

In this paper, a data-driven combined algorithm was proposed for abnormal power loss detection in the distribution network. The proposed algorithm comprised of three stages: abnormal feeder detection, abnormal time detection, and abnormal position detection.

Experiments were conducted using the dataset from the Tianjin Electric Power Company in China. In abnormal feeder detection, the classification algorithm SVM outperformed the three selected clustering algorithms from the comprehensive perspective of the DR and FAR . In abnormal time detection, the control chart correctly detected the abnormal date by analyzing the fluctuation of the r_d . In abnormal position detection, the risk assessment technology was utilized to locate the specific abnormal T_p and to determine the risk level by the proposed rules. The detection results were corresponding with the predefined abnormal scenarios, which suggested that the proposed combined data-driven algorithm could effectively detect abnormal power loss in the distribution network. In further work, more valuable features are needed to extract to improve the detection accuracy of the abnormal feeder type.

REFERENCES

- [1] J. B. Leite and J. R. S. Mantovani, "Detecting and locating non-technical losses in modern distribution networks," *IEEE Trans. Smart Grid*, vol. 9, no. 2, pp. 1023–1032, Mar. 2018.
- [2] S. Zhang, X. Dong, Y. Xing, and Y. Wang, "Analysis of Influencing Factors of Transmission Line Loss Based on GBDT Algorithm," in *Proc. Int. Conf. Commun., Inf. Syst. Comput. Eng. (CISCE)*, Haikou, China, Jul. 2019, pp. 179–182.
- [3] R. Jiang, R. Lu, Y. Wang, J. Luo, C. Shen, and X. Shen, "Energy-theft detection issues for advanced metering infrastructure in smart grid," *Tin-shhua Sci. Technol.*, vol. 19, no. 2, pp. 105–120, Apr. 2014.
- [4] P. Jokar, N. Arianpoo, and V. C. M. Leung, "Electricity theft detection in ami using customers' consumption patterns," *IEEE Trans. Smart Grid*, vol. 7, no. 1, pp. 216–226, Jan. 2016.
- [5] K. Zheng, Q. Chen, Y. Wang, C. Kang, and Q. Xia, "A novel combined data-driven approach for electricity theft detection," *IEEE Trans. Ind. Informat.*, vol. 15, no. 3, pp. 1809–1819, Mar. 2019.
- [6] F. Xiao and Q. Ai, "Electricity theft detection in smart grid using random matrix theory," *IET Gener., Transmiss. Distrib.*, vol. 12, no. 2, pp. 371–378, Jan. 2018.
- [7] S.-C. Huang, Y.-L. Lo, and C.-N. Lu, "Non-technical loss detection using state estimation and analysis of variance," *IEEE Trans. Power Syst.*, vol. 28, no. 3, pp. 2959–2966, Aug. 2013.
- [8] S. Amin, G. A. Schwartz, A. A. Cardenas, and S. S. Sastry, "Game-theoretic models of electricity theft detection in smart utility networks: Providing new capabilities with advanced metering infrastructure," *IEEE Control Syst.*, vol. 35, no. 1, pp. 66–81, Feb. 2015.

- [9] S. Amin, G. A. Schwartz, and H. Tembine, "Incentives and security in electricity distribution networks," in *Decision Game Theory for Security*. Berlin, Germany: Springer, 2012, pp. 264–280.
- [10] Q. Chen, K. Zheng, C. Kang, and F. Huangfu, "Detection methods of abnormal electricity consumption behaviors: Review and prospect," *Automat. Electr. Power Syst.*, vol. 42, no. 17, pp. 189–199, 2018.
- [11] Y. Gao, B. Foggo, and N. Yu, "A physically inspired data-driven model for electricity theft detection with smart meter data," *IEEE Trans. Ind. Informat.*, vol. 15, no. 9, pp. 5076–5088, Sep. 2019.
- [12] M. De Nadai and M. Van Someren, "Short-term anomaly detection in gas consumption through ARIMA and artificial neural network forecast," in *Proc. IEEE Workshop Environ., Energy, Struct. Monitor. Syst. (EESMS)*, Trento, Italy, Jul. 2015, pp. 250–255.
- [13] J. Nagi, K. S. Yap, S. K. Tiong, S. K. Ahmed, and M. Mohamad, "Nontechnical loss detection for metered customers in power utility using support vector machines," *IEEE Trans. Power Del.*, vol. 25, no. 2, pp. 1162–1171, Apr. 2010.
- [14] L. A. Passos Júnior, C. C. Oba Ramos, D. Rodrigues, D. R. Pereira, A. N. De Souza, K. A. P. Da Costa, and J. P. Papa, "Unsupervised non-technical losses identification through optimum-path forest," *Electr. Power Syst. Res.*, vol. 140, pp. 413–423, Nov. 2016.
- [15] L. A. M. Pereira, L. C. S. Afonso, J. P. Papa, Z. A. Vale, C. C. O. Ramos, D. S. Gastaldello, and A. N. Souza, "Multilayer perceptron neural networks training through charged system search and its Application for non-technical losses detection," in *Proc. IEEE PES Conf. Innov. Smart Grid Technol. (ISGT Latin Amer.)*, São Paulo, Brazil, Apr. 2013, pp. 1–6.
- [16] M. Zanetti, E. Jamhour, M. Pellenz, M. Penna, V. Zambenedetti, and I. Chueiri, "A tunable fraud detection system for advanced metering infrastructure using short-lived patterns," *IEEE Trans. Smart Grid*, vol. 10, no. 1, pp. 830–840, Jan. 2019.
- [17] N. F. Avila, G. Figueroa, and C.-C. Chu, "NTL detection in electric distribution systems using the maximal overlap discrete wavelet-packet transform and random undersampling boosting," *IEEE Trans. Power Syst.*, vol. 33, no. 6, pp. 7171–7180, Nov. 2018.
- [18] G. Muruti, F. A. Rahim, and Z.-A. Bin Ibrahim, "A survey on anomalies detection techniques and measurement methods," in *Proc. IEEE Conf. Appl. Inf. Netw. Secur. (AINS)*, Langkawi, Malaysia, Nov. 2018, pp. 81–86.
- [19] M. Z. A. Bhuiyan, G. J. Anders, J. Philhower, and S. Du, "Review of static risk-based security assessment in power system," *IET Cyber-Phys. Syst., Theory Appl.*, vol. 4, no. 3, pp. 233–239, Sep. 2019.
- [20] J. Dongli, D. Yinglong, W. Cunping, and H. Renle, "Research on fault risk ranking and screening of distribution network based on uncertainty theory," in *Proc. China Int. Conf. Electr. Distrib. (CICED)*, Tianjin, China, Sep. 2018, pp. 1561–1565.
- [21] H. Jia, W. Qi, Z. Liu, B. Wang, Y. Zeng, and T. Xu, "Hierarchical risk assessment of transmission system considering the influence of active distribution network," *IEEE Trans. Power Syst.*, vol. 30, no. 2, pp. 1084–1093, Mar. 2015.
- [22] A. Chauhan and S. Rajvanshi, "Non-technical losses in power system: A review," in *Proc. Int. Conf. Power, Energy Control (ICPEC)*, Feb. 2013, pp. 558–561.
- [23] Y. Ren, D. Jiang, T. Xing, and P. Zhu, "Research on software quality control method based on control chart," in *Proc. IEEE 2nd Int. Conf. Comput., Control Ind. Eng.*, Wuhan, China, Aug. 2011, pp. 274–277.
- [24] B. Chen, Y. Yao, C. Tan, Y. Huang, and J. Feng, "A study on spectral characterization and quality detection of direct metal deposition process based on spectral diagnosis," *Int. J. Adv. Manuf. Technol.*, vol. 96, nos. 9–12, pp. 4231–4241, Jun. 2018.
- [25] S. Song, W. Gu, H. Liang, M. Wu, H. Liu, and J. Zhu, "Multiattribute comprehensive risk assessment of distribution network connected with distributed generations," in *Proc. China Int. Conf. Electr. Distrib. (CICED)*, Xi'an, China Aug. 2016, pp. 1–6.
- [26] J. A. Hartigan and M. A. Wong, "Algorithm AS 136: A K-means clustering algorithm," *Appl. Statist.*, vol. 28, no. 1, p. 100, 1979.
- [27] A. Y. Ng, M. I. Jordan, and Y. Weiss, "On spectral clustering: Analysis and an algorithm," in *Proc. Adv. Neural Inf. Process. Syst.*, 2002, pp. 849–856.
- [28] J. D. Banfield and A. E. Raftery, "Model-based Gaussian and non-Gaussian clustering," *Biometrics*, vol. 49, no. 3, p. 803, Sep. 1993.
- [29] C. J. C. Burges, "A tutorial on support vector machines for pattern recognition," *Data Mining Knowl. Discovery*, vol. 2, no. 2, pp. 121–167, 1998.



applied in modeling, optimizing, monitoring the renewable energy systems, and power systems.



HUAN LONG (Member, IEEE) received the B.Eng. degree from the Department of Automation, Huazhong University of Science and Technology, China, in 2013, and the Ph.D. degree in systems engineering and engineering management from the City University of Hong Kong, Hong Kong, in 2017.

She is currently an Assistant Professor with the School of Electrical Engineering, Southeast University. Her research interests include data mining

CHANG CHEN (Student Member, IEEE) was born in Jiangsu, China, in 1996. She received the B.Eng. degree from the Department of Automation, Nanjing University of Aeronautics and Astronautics, China, in 2018. She is currently pursuing the master's degree with the School of Electrical Engineering, Southeast University, China.

Her research interest includes distribution network planning.



WEI GU (Senior Member, IEEE) received the B.S. and Ph.D. degrees in electrical engineering from Southeast University, China, in 2001 and 2006, respectively.

From 2009 to 2010, he was a Visiting Scholar with the Department of Electrical Engineering, Arizona State University, Tempe, AZ, USA. He is currently a Professor with the School of Electrical Engineering, Southeast University. He is the Director of the Institute of Distributed Generations

and Active Distribution Networks. His research interests include distributed generations and microgrids, and integrated energy systems.



JIHUA XIE received the B.S. degree in electrical engineering from Southeast University, China, in 1990. He is currently an Associate Professor with the School of Electrical Engineering, Southeast University. His research interests include the cyber security of smart grid, the Internet of Things security, and industrial control systems.



ZHENG WANG received the M.S. degree in electrical engineering from North China Electric Power University, Baoding, China, in 2008. She is currently with the State Grid Tianjin Electric Power Research Institute. Her research interest includes power system analysis.



GUODONG LI received the M.S. degree in electrical engineering from Tianjin University, Tianjin, China, in 2010. He is currently with the State Grid Tianjin Electric Power Research Institute. His research interest includes power quality analysis.

• • •